



Efficient Removal of Congo Red from Wastewater Using Date-Pit Activated Carbon: Isotherm, Kinetic, and Thermodynamic Studies

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Abstract

The presence of dye molecules in water leads to numerous adverse effects on both human health and aquatic life. This work explores the valorization of agricultural waste through the synthesis and functionalization of date pit-based activated carbon (DP-AC) derived from *Phoenix dactylifera* (Ghars variety) pits collected at three distinct developmental stages. The base DP-AC was prepared via chemical activation using phosphoric acid (H_3PO_4), with the objective of enhancing its adsorption performance. The synthesized adsorbent was comprehensively characterized using N_2 adsorption–desorption isotherm, FTIR, XRD, SEM, TGA, pH_{PZC} , and Boehm titration. The BET analysis revealed that DP-AC possesses a microporous structure with a surface area of $946.68 \text{ m}^2/\text{g}$ and a total pore volume of $0.46 \text{ cm}^3/\text{g}$. Moreover, SEM images revealed that DP-AC exhibited noticeable structural modifications, including improved surface texture and pore accessibility. These modifications translated into superior adsorption behavior. Batch adsorption experiments were conducted to evaluate the effect of pH, contact time, adsorbent dose, dye concentration, and temperature on the removal efficiency of Congo Red dye. DP-AC exhibited an equilibrium time of 120 min, with a maximum adsorption capacity of 156.04 mg/g under optimal conditions. Isotherm modeling revealed that the Langmuir model best described the adsorption process, indicating monolayer adsorption on a homogeneous surface. Kinetic analysis showed that the data fit well with the pseudo-second-order model, highlighting surface-controlled adsorption mechanisms. Kinetic analysis revealed a good agreement with the pseudo-second-order model, indicating that the adsorption process was predominantly governed by surface-controlled mechanisms. Thermodynamic parameters (ΔG° , ΔH° , and ΔS°) confirmed that the adsorption onto AC was spontaneous and endothermic in nature.

Keywords Activated carbon · Adsorption · Congo Red · Date pits · Kinetics · Wastewater

1 Introduction

Pollution is defined as the environmental degradation caused by human-induced substances, resulting in the loss of biodiversity, the emergence of new environmental challenges,

and negative impacts on human health, notably contributing to the acceleration of global warming [1, 2]. The impacts of this pollution extend beyond air and soil, significantly affecting a substantial portion of water resources as well [3]. Although water covers over 70% of the Earth's surface, only about 3% is fresh and suitable for human consumption, while the remaining 97% consists of saline water [4]. Globally, nearly 4 billion individuals experience water scarcity for a minimum of 1 month each year. It is estimated that up to 4 billion individuals suffer from water shortages for at least a month on an annual basis [5].

Dyes are widely employed across various industrial sectors, including textiles [6], paper [7], leather [8], food [9], and cosmetic industries [10]. Moreover, these industries require substantial amounts of water in their operations. The persistence and degradation of dyes in wastewater can substantially affect water quality and have detrimental effects on human health [11]. For example, in the textile industry,

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approximately 10% of the dyes used are lost during the dyeing process, with about 2% directly released into the environment as untreated aqueous effluents. These dyes maintain their structural integrity and coloration when exposed to soil, sunlight, microbial activity, and perspiration, and they exhibit considerable resistance to biodegradation during conventional wastewater treatment processes [12]. Therefore, dyes must be removed before industrial wastewater is released into water bodies.

To diminish the impact of this pollution, various techniques have been developed and tested for the treatment of dye-containing effluents, including biological processes [13], coagulation/flocculation [14], photodegradation [15], ozonation [16], electrochemical methods [17], advanced oxidation process [18], and membrane separation [19]. Adsorption is unique among the available treatment techniques because of its high efficacy, affordability, ease of use, and broad range of applications [20]. Extensive research has long been conducted on the removal of dyes from aqueous solutions using a variety of materials, especially activated carbons (ACs) [21]. In recent years, a wide range of adsorbents has been explored for the removal of dyes including clay [22], zeolite [23], layered double hydroxides [24], metal organic framework [25], silicon nitride [26], chitosan [27], and hydroxyapatite [28].

The high cost of ACs highlights the need to identify cheaper, effective, and naturally abundant adsorbents. Any inexpensive biomass characterized by elevated carbon content and a reduced inorganic fraction may be utilized as a feedstock for ACs preparation [29]. An environmentally sustainable approach involves converting waste materials into ACs for use in pollution control applications. One of the key challenges in commercial AC production is identifying inexpensive precursors. In this context, extensive research has focused on producing ACs from industrial and agricultural wastes, such as apricot stones [30], olive stones [31], jujube stones [32], date stones [33], *Prunus persica* seed stones [34], cherry stones [35], ornamental stone [36], etc.

The removal of Congo Red (CR) from aqueous solutions using ACs derived from various precursors has been extensively studied. For instance, M. Husaini et al. [37] utilized almond seed shells to produce AC as a low-cost adsorbent for CR removal. Sujesh Sudarsan et al. [38] studied the removal of CR from aqueous solutions using AC derived from *Spathodea campanulata* flowers and reported a maximum adsorption capacity of 59.27 mg/g. A. Wang et al. [39] investigated the adsorption mechanisms of surface-modified AC incorporating bacteria and reported a maximum adsorption capacity of 234.6 mg/g at 25 °C. Hao et al. [40] utilized waste newspaper as a cellulose-based precursor to produce activated carbon using sodium salts for Congo Red removal, achieving an adsorption capacity of 299 mg/g. Djaber et al. [41] synthesized ACs from date palm spathes via phosphoric

acid activation at 450 °C and reported a remarkable adsorption capacity of 1032 mg/g for CR. From the above studies, it is evident that adsorption is a highly effective approach for CR removal; however, the reported adsorption capacities differ widely depending on the precursor and activation method. This variation highlights the need to develop new, low-cost, and sustainable adsorbents with competitive performance from abundant waste materials, a research area that is not widely explored and still limited in the literature.

Algeria produces approximately 1.058 million tons of dates annually, encompassing more than 400 distinct varieties [42]. Date pit (DP) account for roughly 10% of the total fruit weight, and this lignocellulose-based agricultural by-product consequently represents a significant solid environmental pollutant. Hence, DPs, which is proved to be an environmentally friendly and low-cost adsorbent, were purposefully used in this study to prepare DP-AC due to their abundance as agricultural waste and their potential as superior precursors for creating adsorbents with advantageous structural, morphological, and chemical characteristics. The proposed synthesis route offers a low-cost, sustainable approach for producing efficient adsorbents, thus contributing to the valorization of local biomass resources and representing a meaningful advancement over similar studies.

Physical and chemical activation techniques, or their combination, can be used to prepare AC. Since it has several benefits, including single-step treatment, simplicity, low operating costs, higher yields, and improved porous structure, chemical activation is the most widely used method. We describe the production of low-cost ACs from agricultural wastes using H_3PO_4 and its use for the removal of CR. Batch adsorption was used to examine the effects of temperature, pH, adsorbent dose, contact time, and initial CR concentration. In addition, the adsorption's equilibrium, kinetics, and thermodynamics were investigated. The prepared DP-AC is expected to significantly contribute to wastewater treatment technologies and holds strong potential for large-scale applications. Thus, widespread and great progress in this area can be expected in the future.

2 Experimental

2.1 Biomass

Date pits (DP) at three distinct growth stages of *Phoenix dactylifera* (Ghars variety) were collected from the Ouargla region in southeastern Algeria: Kamri stage (May–June), Rutab stage (July–August), and Ghars stage (September). The DP were utilized for the preparation of AC, which was subsequently employed for the removal of CR dye from aqueous solutions.

2.2 Preparation of DP-AC

DP were first thoroughly washed with hot water several times (each variety separately) to remove any impurities, then sun-dried for 20 days. Once fully dried, they were crushed and sieved together to obtain a uniform particle size of 1 mm. For chemical activation, 10 g of DP was mixed with phosphoric acid (H_3PO_4) at a 1:1 weight ratio (g/g), and the mixture was left to dry at 105 °C for 24 h. The dried samples were then carbonized in a horizontal tube furnace at 700 °C for 2 h under a nitrogen atmosphere (150 mL/min) with a heating rate of 5 °C/min. Following the cooling step, the carbonized materials were thoroughly washed with distilled water until a neutral pH was reached, then filtered and dried at 105 °C for 10 h. Subsequently, 90% of each batch was pooled to produce a single, representative AC sample, herein referred to as DP-AC. This procedure was adapted from previously reported methods with slight modifications [43, 44].

2.3 Characterization Techniques

A range of characterization techniques was employed to thoroughly assess the physicochemical properties of the synthesized adsorbents. Thermogravimetric analysis (TGA) was performed, using Setaram LABSYS evo, on both raw and activated samples under a nitrogen atmosphere, with a heating rate of 10 °C/min in the temperature range of 30–900 °C. The porous structure of the DP-AC was analyzed using N_2 adsorption–desorption isotherms at 77 K after degassing at 150 °C for 12 h using a Micrometrics equipment (AZAP 2020 Plus). The BET method was used to determine the specific surface area, while the t-plot method provided the micropore surface area and volume. The total pore volume was derived from nitrogen adsorption at a relative pressure of 0.998. The surface morphology of the samples was analyzed using an EVO 15 SEM scanning electron microscope (SEM). The elemental composition was identified using energy-dispersive X-ray spectroscopy (EDS) combined with environmental scanning electron microscopy (ESEM). Powder X-ray diffraction (XRD) patterns were recorded between $2\theta = 10^\circ$ and 50° on an OLYMPUS BTX-716 X-ray diffractometer with CuK_α radiation source. Fourier transform infrared spectroscopy (FTIR) was employed to identify the functional groups on the adsorbent surface, using a Shimadzu 8300-S instrument over a spectral range of 400–4000 cm^{-1} . The point of zero charge (pH_{PZC}) was measured using the pH drift technique [45]. Flasks containing 10 mL of 0.1M KNO_3 at initial pH values (2–12) were adjusted with 0.1M HCl or NaOH, then mixed with 10 mg of DP-AC and agitated for 180 min at room temperature. The pH_{PZC} was determined from the intercept between $\text{pH}_f - \text{pH}_i$ and pH_i where pH_i and pH_f are initial and final pH values, respectively. The pH at

which the curve intersects the axis bisector is considered as the pH_{PZC} of the bio-material. The acidic and basic surface functionalities were assessed using Boehm titration following standard protocols [1].

2.4 Materials and Batch Experimentation

Adsorption experiments of CR were performed using the DP-AC material to investigate the isotherm models, thermodynamic parameters, and kinetic mechanisms governing the adsorption process. All experiments were conducted in triplicate to ensure reproducibility, with the relative standard deviation maintained within $\pm 2\%$. A standard stock solution of CR at a concentration of 1000 mg/L was prepared by dissolving the appropriate amount of dye in 1 l of distilled and deionized water. Working solutions were prepared through successive dilutions of the stock solution. The pH of each solution was adjusted according to the requirements of each batch experiment.

The adsorption experiments took place in flasks that underwent gentle agitation on a temperature-controlled water-bath shaker. In each trial, 10.0 mL of CR solution, varying in concentrations, was introduced to a measured quantity of DP-AC within a 50 mL Erlenmeyer flask. The mixtures were agitated at a constant shaking speed using a thermostatic shaker for predetermined time intervals. Upon completion of the adsorption, the suspension was gathered via centrifugation at 8000 rpm/min for 10 min. Assessment of the CR residual concentrations was conducted using a Hach DR 6000 UV–Vis spectrophotometer at a wavelength of 498 nm.

The adsorbed amount of CR onto the DP-AC, Q_e (mg/g), and removal percentage (η) were calculated using following equations:

$$Q_e = \frac{V}{m} (C_0 - C_e), \quad (1)$$

$$\eta = \frac{(C_0 - C_e)}{C_0}, \quad (2)$$

where C_0 (mg/L) is the initial CR concentration, C_e (mg/L) is the equilibrium CR concentration, m (g) is the mass of the adsorbent, and V (L) is the volume of the working solution.

The kinetics of CR dye adsorption was investigated by agitating a series of conical flasks containing a fixed mass of adsorbent with 10 mL of dye solution (25 mg/g). The experiments were conducted at a controlled temperature over contact times ranging from 0 to 270 min. In addition, the influences of adsorbent dose (0.2–8 g/L), pH (2–12), initial concentration (10–200 mg/L), and temperature variations (25, 35, and 55 °C) were examined.

2.5 Adsorption Models

To determine the most suitable model for describing the experimental data of CR adsorption onto DP-AC, the kinetic mechanism was examined by applying the pseudo-first (PFO) [46], pseudo-second (PSO) [47], and intra-particle diffusion (IPD) models [48] to the results. Furthermore, the experimental isotherm data were evaluated to determine the distribution of the adsorbate between the solid and liquid phases at different equilibrium concentrations, with the aim of gaining deeper insight into the adsorption process. Consequently, the Freundlich [49], Langmuir [50], and Temkin [51] models were applied to evaluate the dye adsorption parameters onto DP-AC. Further analysis of the Langmuir equation involves the use of a dimensionless separation factor, R_L , defined as [52]:

$$R_L = \frac{1}{1 + K_L C_0}. \quad (3)$$

The R_L values indicate whether the process is spontaneous ($0 < R_L < 1$), unfavorable ($R_L > 1$), linear ($R_L = 1$), or irreversible ($R_L = 0$).

The quality of the experimental data fitting to these models was judged based on the assessment of the correlation coefficients (R^2) and also the absolute relative error (ARE) as given below [53]:

$$ARE_{\text{kinetic}} = \frac{\sum \left| \frac{Q_{t,\text{exp}} - Q_{t,\text{calc}}}{Q_{t,\text{exp}}} \right|}{N} \times 100 \quad (4)$$

$$ARE_{\text{isotherm}} = \frac{\sum \left| \frac{Q_{e,\text{exp}} - Q_{e,\text{calc}}}{Q_{e,\text{exp}}} \right|}{N} \times 100 \quad (5)$$

where $Q_{t,\text{exp}}$, $Q_{t,\text{calc}}$; and $Q_{e,\text{exp}}$, and $Q_{e,\text{calc}}$ are the measured and calculated retained CR amounts at given time t and at saturation, respectively. N denotes the number of experiments runs.

3 Results and Discussion

3.1 Characterization of DP-AC

3.1.1 TGA

TGA is a key technique for material characterization, providing an estimate of sample weight loss as a function of temperature. The peak of the first derivative (DTA) indicates the inflection point, corresponding to the maximum rate of change on the weight loss curve. Figure 1a and b presents

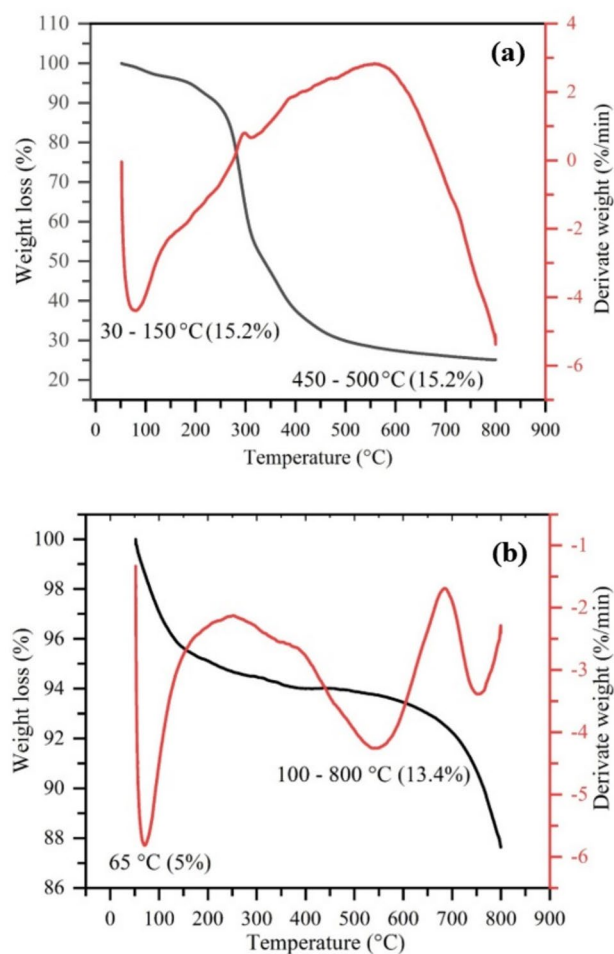


Fig. 1 TGA/DTA graphs of DP (a) and DP-AC (b)

the TGA and DTA curves for both raw biomass (DP) and the DP-AC. Regarding the DP material, the first decomposition stage occurred between 30 and 150 °C, where a weight loss of 15.2% was observed likely due to the evaporation of moisture physically adsorbed on the surface. A smaller but distinct loss occurs between 450 and 550 °C (10%), which corresponds to the breakdown of natural components such as hemicellulose, cellulose, and lignin consistent with findings by Mohammed and Adewole [54]. This stage of decomposition (450–550 °C) is mainly associated with the degradation of char, which results from the breakdown of biomass components, leaving behind ash and char residues [55]. At temperatures above 600 °C, the material appears to reach stabilization, demonstrating good thermal stability. Compared to biochars, biomass undergoes thermal decomposition earlier due to its lower thermal stability.

On the other hand, the DP-AC material shows a first loss at around 50 °C, which was attributed to the release of physically adsorbed moisture and volatile compounds. The minor endothermic peak detected around 65 °C, associated with

approximately 5% weight loss, is attributed to the desorption of physically adsorbed moisture rather than true thermal degradation. The subsequent stage exhibited a slower weight loss rate, with only a 13.4% reduction occurring over a broad temperature range from 100 to 800 °C. In this stage, most of the weight loss resulted from the decomposition of hemicellulose and cellulose, whereas at temperatures above approximately 450 °C, it was primarily attributed to lignin degradation. At 800 °C, the ash residue amounted to 87%, indicating the high thermal stability of the material (DP-AC) across a wide temperature range.

3.1.2 XRD

XRD patterns provide valuable information, including the identification of crystal phases and their composition, preferred crystallographic orientations, and the detection of unknown elements [56]. The XRD profile of the prepared DP-AC sample, presented in Fig. 2, illustrates the characteristic diffraction peaks that provide insights into its structural and crystalline properties. Two broad diffraction peaks, detected at around $2\theta = 25^\circ$ and 43° , correspond to the (002) and (100) planes, respectively, which are characteristic of amorphous carbon. The broad peak near $2\theta = 25^\circ$, corresponding to the (002) plane of turbostratic carbon, indicates the presence of an amorphous structure—an advantageous feature for producing well-defined porous adsorbents [57]. This feature suggests the presence of a partially ordered graphitic structure, a typical characteristic of many ACs [58, 59]. In addition, a peak near 17° may indicate some residual cellulose or incomplete carbonization of the original biomass.

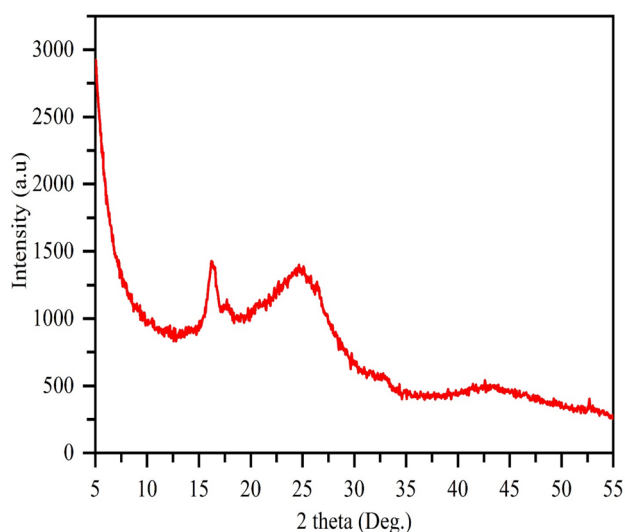


Fig. 2 XRD pattern of the DP-AC

3.1.3 N₂ Adsorption–Desorption Isotherm

The N₂ adsorption–desorption isotherm of DP-AC (Fig. 3) exhibits combined features of type I and type II isotherms according to the IUPAC classification. The presence of an H3 hysteresis loop at relative pressures above 0.4 indicates capillary condensation within narrow, slit-shaped mesopores, confirming the coexistence of microporous and mesoporous structures in the material [60]. The textural characteristics of the prepared DP-AC material were evaluated using N₂ isotherm analysis. The DP-AC adsorbent showed a BET surface area (SSA) of 946.68 m²/g, with a total pore volume of 0.46 cm³/g, of which approximately 0.18 cm³/g corresponds to micropores. The microporous surface area was 863.08 m²/g accounting for about 91% of the total surface area, which is consistent with the observed isotherm profile. This SSA is notably higher than those reported for ACs prepared from other biomass sources, including agricultural wastes (289.86 m²/g) [61], cashew nut shell (407.80 m²/g) [62], peanut shells (526.34 m²/g) [63], and industrial food waste (822 m²/g) [64]. These parameters of the DP-AC textural properties would improve dyes' removal efficiency from aqueous solutions.

3.1.4 SEM

The SEM images reveal the particle growth, surface morphology, as well as the shape and size of the particles, allowing for the investigation of their impact on conductivity, and on both electrical and optical properties [65]. Furthermore, SEM analysis provides valuable insight into the evolution of the adsorbent's surface characteristics. Figure 4 shows the surface morphology of the DP-AC prepared from DPs. The SEM image shows a highly porous and uneven surface,

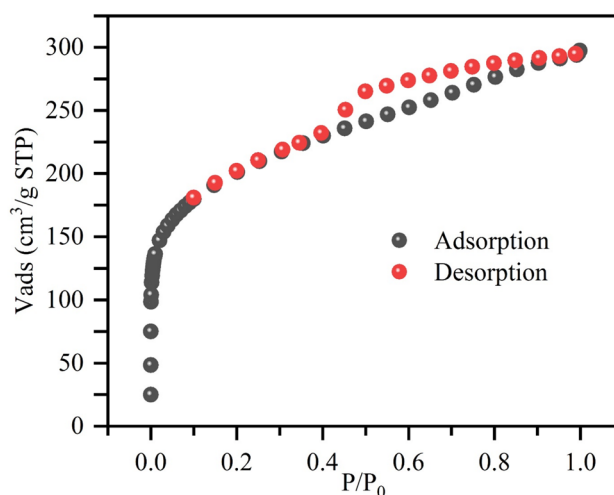


Fig. 3 N₂ adsorption–desorption isotherms at 77 K of the DP-AC

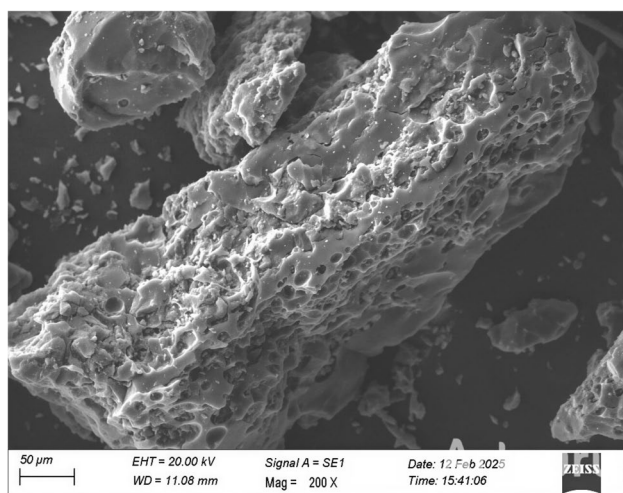


Fig. 4 SEM micrograph of the prepared DP-AC

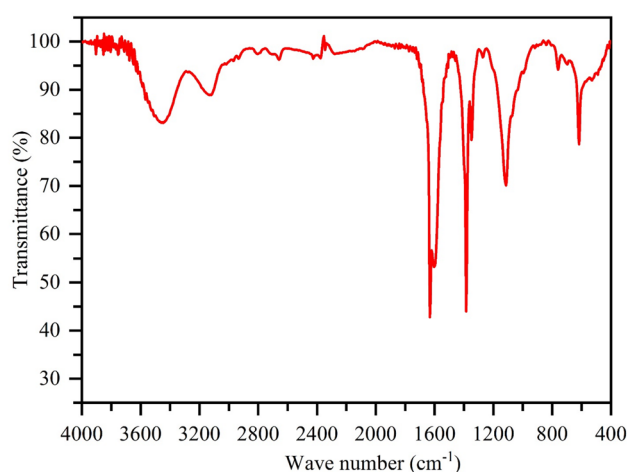


Fig. 5 FTIR spectrum of the prepared DP-AC

marked by irregular cavities and interconnected channels. This morphological disorder aligns well with the XRD results. Together, these observations point to an overall amorphous structure with traces of local graphitic order—a combination that is actually favorable for adsorption, as it offers high surface area and good pore accessibility [3].

3.1.5 FTIR

Infrared spectroscopy is a powerful technique used to identify molecules, particularly by detecting and characterizing their functional groups [66, 67]. Figure 5 presents the FTIR spectra of the DP-AC sample obtained using both physical and chemical activation with H_3PO_4 . The FTIR spectrum of the DP-AC displays a broad O–H stretching band around 3400 cm^{-1} , indicating the presence of hydroxyl groups on

the surface. Two noticeable peaks appear near 1600 and 1650 cm^{-1} , which can be attributed to aromatic $\text{C}=\text{C}$ and carbonyl $\text{C}=\text{O}$ vibrations, respectively. Additional absorption bands between 1000 and 1200 cm^{-1} suggest the presence of $\text{C}-\text{O}$ and $\text{P}-\text{O}$ functional groups, likely introduced during the phosphoric acid activation process [68].

The detection of these oxygen-containing groups suggests that the surface of the DP-AC is mainly acidic. This was confirmed by Boehm titration, which quantified the surface acidity and revealed the presence of carboxylic (0.424 mmol/g), lactonics (0.600 mmol/g), and phenolic (0.043 mmol/g) groups, resulting in a total acidity of 1.067 mmol/g . On the other hand, the total basicity was measured at 0.573 mmol/g , reinforcing the dominance of acidic sites. This acidity is further supported by the pH_{PZC} , found to be 5.5, a value below neutral pH, confirming the net acidic nature of the surface [69]. Such a surface chemistry plays an important role in enhancing the interaction with anionic dyes like CR, mainly through electrostatic attraction and hydrogen bonding mechanisms [70].

The FTIR, SEM/EDX, and BET analyses confirm that H_3PO_4 -activated DP (DP-AC) possesses excellent structural and textural characteristics. The high surface area, well-developed microporosity, and abundance of surface functional groups collectively enhance their adsorption performance. These results demonstrate the potential of DP as a sustainable and efficient precursor for producing high-quality AC suitable for various adsorption applications.

3.2 Adsorption Studies

3.2.1 Effect of Operational Variables on Dye Adsorption

The change in the amount of CR adsorbed onto DP-AC as a function of contact time is illustrated in Fig. 6. Adsorption efficiency increases rapidly at first reaching 72 mg/g after just 30 min, then more moderately. The increasing trend stopped when equilibrium was reached, within 200 min. This rapid initial uptake is attributed to the high density of accessible binding sites on the carbon surface [71]. Over time, the rate of adsorption slowed down, likely due to the gradual saturation of those sites and the growing influence of intra-particle diffusion resistance. Based on these findings, a contact time of 120 min was selected as the optimal equilibrium time for the remaining experiments.

One of the key factors influencing sorption capacity is the amount of sorbent present in the liquid phase, as it determines the adsorption capacity for a given initial dye concentration [72]. Determining the optimal dosage of DP-AC adsorbent in the adsorption process is crucial, as it directly impacts the overall application cost [73]. The quantity of adsorbent used is a critical parameter that defines its ability to remove a specific amount of adsorbate from a solution

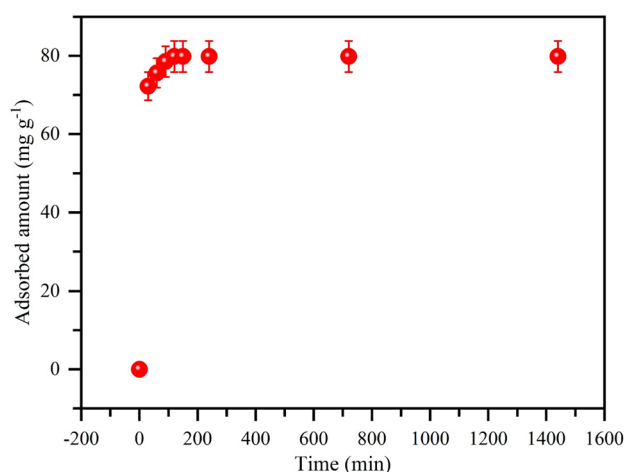


Fig. 6 Effect of contact time on the removal efficiency of DP-AC towards CR

under given conditions [74]. To determine the optimal adsorbent mass for maximum removal efficiency, the influence of DP-AC dosage on CR adsorption was investigated at a constant adsorbate concentration. As shown in Fig. 7, the dye removal percentage increased with higher adsorbent dosages; for example, it rose from 85.18 to 99.93% as the dosage increased from 0.2 to 2 g/L. When the adsorbent dosage exceeded 2 g/L, the dye removal efficiency remained constant at 99.93%, suggesting that nearly complete adsorption was achieved with 2 g/L of adsorbent at a dye concentration of 25 mg/L. This improvement can be explained by the greater number of available active sites on the carbon surface, which enhances dye adsorption [75]. However, beyond 2 g/L, the removal efficiency showed only minor changes, suggesting that increasing the dose further does not significantly impact the outcome. Based on these results, 2 g/L was

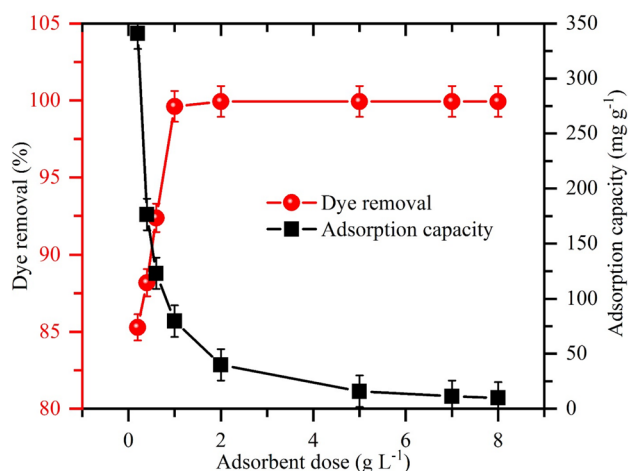


Fig. 7 Effect of adsorbent dose on the CR removal over DP-AC

selected as the optimal dose for subsequent experiments. Moreover, Fig. 6 clearly shows that the decrease in sorption capacity with increasing adsorbent dosage, at constant dye concentration and volume, may be attributed to the saturation of adsorption sites caused by particulate interactions such as aggregation [76].

In adsorption studies, pH optimization plays a crucial role in environmental protection, as variations in pH can alter the charges of both adsorbent and adsorbate molecules [77]. In this order, the effect of pH on the adsorption of CR was investigated by varying the pH of the dye solution from 2 to 12, while keeping other parameters constant: dye concentration = 80 mg/L, adsorbent dose = 2 g/L, and contact time = 120 min. The results corresponding to the variation in pH are presented in Fig. 8.

As shown in Fig. 7, the adsorption efficiency reached its maximum at pH = 2, then gradually declined as the pH increased. This behavior can be understood by considering the surface charge characteristics of the DP-AC, which has a pH_{PZC} of 5.5 (Fig. 9). At pH values below 5.5, the surface carries a positive charge, which promotes electrostatic attraction with negatively charged dye molecules like CR. However, as the pH rises above the pH_{PZC} , the surface becomes negatively charged, leading to repulsion between the surface and the anionic sulfonate groups ($-SO_3^-$) of the dye. This repulsion reduces the interaction between the dye and the adsorbent, and consequently lowers the adsorption efficiency [78, 79].

Interestingly, a slight increase in adsorption was observed at pH = 8, deviating from the general trend of decreasing efficiency with increasing pH. This unexpected behavior may be related to the involvement of other non-electrostatic mechanisms, such as hydrogen bonding between the dye molecules and neutral functional groups present on the carbon surface at near-neutral pH [80].

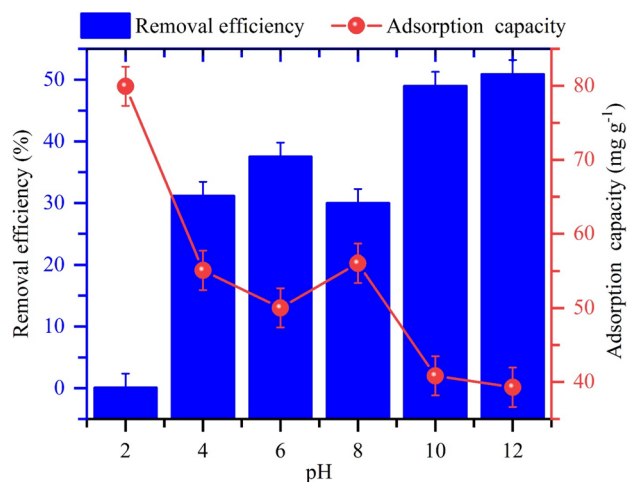


Fig. 8 Effect of pH on the adsorption capacity of DP-AC

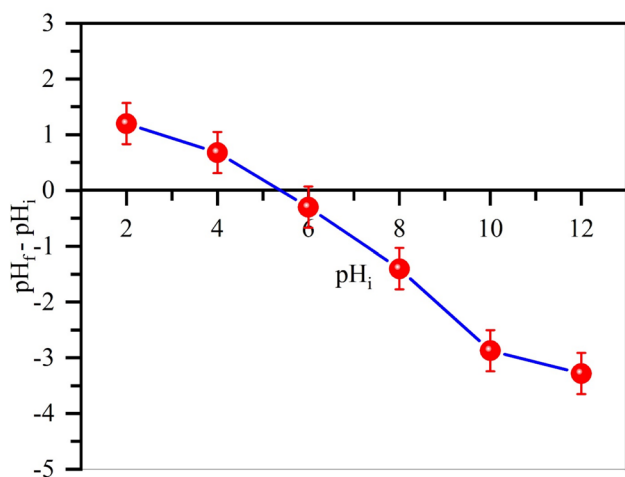


Fig. 9 Point of zero charge of the DP-AC

These results suggest that while electrostatic forces are dominant at low pH, alternative interactions can still contribute to adsorption under specific conditions. This observation is consistent with findings reported for other bio-based adsorbents [81].

3.2.2 Adsorption Isotherm Study

The adsorption isotherm delineates the thermodynamic equilibrium existing between an adsorbent and an adsorbate, serving as a valuable tool in comprehending the adsorption mechanism [82]. Examining the equilibrium adsorption process holds significance as it furnishes crucial insights into factors such as affinity, binding energy between the adsorbate and adsorbent, and the uptake capacity. Furthermore, it aids in making inferences regarding the adsorption mode, whether it involves a monolayer or multilayer fixation mechanism, and the potential presence of lateral interactions among molecules [25]. The investigation into the CR sorption was conducted using a constant weight of the prepared DP-AC (2 g/L) at ambient temperature. The resulting adsorption isotherms were obtained following a 4-h adsorption period, as illustrated in Fig. 10.

The adsorption isotherm obtained in this case (Fig. 10) was compared to models of adsorption isotherms from the literature [83]. It can be observed that the adsorption isotherm of the amount of dye increased rapidly for low concentrations and then stabilized to reach a plateau corresponding to saturation of the adsorption sites. This is a type L isotherm, indicating a monolayer adsorption and low competition between the CR and solvent molecules.

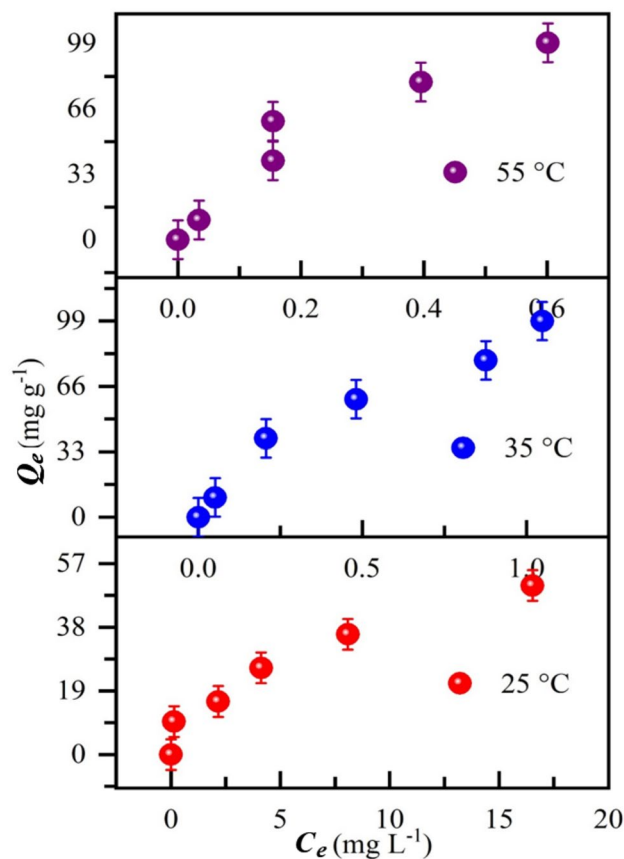


Fig. 10 Adsorption equilibrium data of CR removal over DP-AC

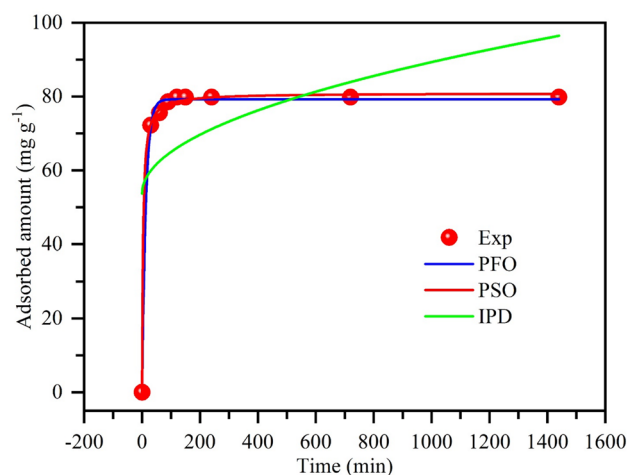
3.2.3 Kinetic Data Modeling

Adsorption kinetics examines the rate at which adsorption occurs, serving as an essential criterion for selecting an appropriate adsorbent [84]. A kinetic study was conducted to elucidate the adsorption reaction mechanisms, with particular emphasis on reaction order and rate constants. The mechanism governing CR adsorption onto DP-AC was investigated by fitting the experimental data to the PFO, PSO, and IPD kinetic models. The corresponding kinetic parameters are summarized in Table 1 and illustrated in Fig. 11. Kinetic modeling of the experimental data using the PFO and PSO equations revealed that the PSO model offered a superior fit. This conclusion is supported by its higher R^2 value (0.991), lower ARE between experimental and theoretical curves (0.76%), and a calculated equilibrium adsorption capacity closely matching the experimental value (Table 1). This indicates that the adsorption process is mainly controlled by interactions at the adsorbent surface.

The superior fit of the pseudo-second-order model indicates that the adsorption rate is primarily governed by the availability of active sites on the DP-AC surface. Although this model is often linked to chemisorption, previous studies

Table 1 Kinetics parameters deduced from the non-linear models for CR adsorption on DP-AC

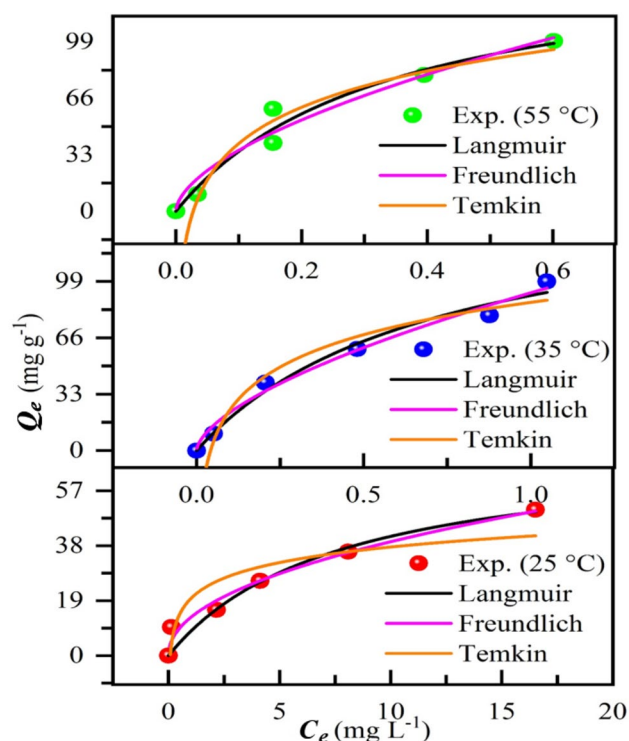
	Parameters	Values
Pseudo-first order	$Q_{e,exp}$	79.8
	k_1 (min ⁻¹)	0.078
	Q_e (mg/g)	79.2
	R^2	0.998
	ARE (%)	1.16
Pseudo-second order	k_2 (g/mg/min)	0.0035
	Q_e (mg/g)	80.89
	R^2	0.999
	ARE (%)	0.76
	k_{diff}	1.125
Intra-particle diffusion	C	53.71
	R^2	0.249
	ARE (%)	14.99

**Fig. 11** Kinetic data modeling

have demonstrated that it can also effectively describe surface-controlled physisorption, particularly for adsorbents with heterogeneous functional groups or variable surface energies [85, 86]. Hence, the adsorption mechanism in this work likely involves both physical and weak chemical interactions, with surface processes predominantly controlling the overall adsorption rate.

3.2.4 Isotherm Data Modeling

Adsorption isotherms provide valuable insights into adsorbate–adsorbent interactions, which are essential for optimizing adsorbent utilization [20]. The equilibrium data for MG and CR adsorption onto DP-AC were fitted to the Langmuir, Freundlich, and Temkin isotherm models, with the results presented in Fig. 12 and summarized in

**Fig. 12** Isotherm data modeling**Table 2** Isotherm parameters deduced from the non-linear models for CR adsorption on DP-AC

Parameters		Values		
		25 °C	35 °C	55 °C
Langmuir	K_L	0.131	1.392	2.946
	R_L	0.433	0.067	0.032
	Q_{max} (mg/g)	72.76	156.04	153.44
	R^2	0.954	0.984	0.969
	ARE (%)	19.06	7.50	17.37
Freundlich	K_F	13.21	92.34	136.46
	$1/n$	2.11	1.62	1.72
	R^2	0.968	0.979	0.934
	ARE (%)	13.96	14.84	26.46
Temkin	B	7.75	27.31	30.64
	A	12.67	23.96	36.32
	R^2	0.862	0.970	0.966
	ARE (%)	30.97	15.99	16.73

Table 2. Among these, the Langmuir model showed the best fit to the experimental data, with the higher correlation coefficient (R^2) and lower ARE (Table 2). That is to say the mechanism applied corresponds to a monolayer adsorption which involves identical, independent, and limited adsorption sites. Moreover, the Langmuir model

provided favorable R_L values ($(0 < R_L < 1)$), pointing to the potential for monolayer adsorption on some uniform sites.

On the other hand, the Freundlich isotherm model stipulates a heterogeneous adsorbent surface covered with multiple layers of the solute. The $1/n$ parameter in the Freundlich isotherm known as adsorption intensity (heterogeneity factor), which indicates the adsorption intensity of CR dye on the applied adsorbents, had values below unity (from 0.474 to 0.616), suggesting a convex shape for the isotherms. This implies that significant adsorption can occur even at high dye concentrations. Furthermore, the Temkin isotherm characterizes the interaction between adsorbent and adsorbate species, assuming that the heat of sorption decreases linearly with increasing surface coverage.

The adsorption of CR onto DP-AC followed a multifaceted mechanism involving both physical and chemical interactions. Initially, the large surface area and mesoporous structure promoted rapid uptake through physisorption, while the pseudo-second-order model indicated increasing chemisorption influence as equilibrium was approached. Functional groups such as $-\text{OH}$, $\text{C}=\text{O}$, and aromatic structures further facilitated hydrogen bonding, π - π stacking, and electrostatic interactions, collectively enhancing adsorption efficiency across varying pH conditions. Moreover, the superior fit of the Langmuir model supports significant monolayer adsorption on energetically uniform sites, consistent with the chemisorptive contribution. A similar observation was reported in previous studies, where CR dye adsorption onto *Spathodea campanulata* flowers AC [38].

The Langmuir isotherm model was employed to evaluate the maximum adsorption capacity (Q_{max}), reflecting the complete monolayer coverage of the adsorbent surface. The Q_{max} representing the sorption capacity at full monolayer coverage, was determined to be 72.76, 156.04, and 153.44 mg/g at temperatures of 25, 35, and 55 °C, respectively. To assess the efficiency of DP-AC in removing CR, its performance was compared with that of other reported adsorbents.

Table 3 presents a comparative analysis of various ACs reported for CR removal. The DP-AC developed in this study demonstrates a markedly higher surface area and pore volume than many bio-based ACs, contributing to its enhanced adsorption performance. Its maximum adsorption capacity ranks third among the 11 adsorbents evaluated, highlighting its competitive efficiency and promising potential. These results underscore the unique combination of structural and surface properties of DP-AC, which enable effective CR removal compared to other reported ACs.

3.3 Thermodynamic Parameters Analysis

The thermodynamic parameters for the adsorption of CR onto the prepared DP-AC (Table 4) were calculated using the following equations [87]:

$$\Delta G^\circ = -RT \ln K_L,$$

$$\ln K_L = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT}.$$

The calculated Gibbs free energy values (ΔG°) at 298, 308, and 328 K were all negative, confirming that the adsorption process is spontaneous across the tested temperature range [49]. The slightly negative ΔG° values (-1 to -10 kJ/mol) indicate that adsorption proceeds mainly through weak physisorption [88]. However, the reduction in the absolute magnitude of ΔG° with increasing

Table 4 Thermodynamic parameters

Temperature (K)	ΔG° (kJ/mol)	ΔH° (kJ/mol)	ΔS° (kJ/mol/K)
298	-3.43	35.327	0.111
308	-0.867		
328	-0.386		

Table 3 Comparative study on the adsorptive removal of CR by different activated carbons materials

Biomass	Activation method	Adsorbate	Q_{max} (mg/g)	Reference
Corn stalk biochar	Microwave + H_3PO_4 activation	CR	124	[92]
Bombax buonopozense	H_3PO_4 activation	CR	510.2	[93]
<i>Cymbopogon winterianus</i>	H_3PO_4 activation	CR	6.25	[94]
<i>Hevea brasiliensis</i> seed shells	H_3PO_4 activation	CR	55.87	[95]
<i>Arachis hypogaea</i> shell	ZnCl_2 activation	CR	21.9	[96]
<i>Hevea brasiliensis</i> seed shells	NaCl activation	CR	50.51	[95]
Date pits	H_3PO_4 activation	CR	156.04	This work
Edamame shell	KOH activation	CR	1509.60	[97]
Water hyacinth leaf and stem	Thermal activation	CR	13.9	[98]
<i>Spathodea campanulata</i> flowers	H_3PO_4 activation	CR	59.27	[38]
Waste black cardamom peels	KOH activation + carbonization	CR	69.93	[99]

temperature suggests that spontaneity decreases slightly at higher temperatures, likely due to the weak physical interactions governing the process. The positive enthalpy change ($\Delta H^\circ = 35.32$ kJ/mol) confirms that the adsorption process is endothermic, meaning that increased temperature enhanced the adsorption capacity. Such moderately positive (ΔH° values (20–40 kJ/mol) indicate that the absorbed energy is mainly used to remove surface-bound water or reorganize active sites rather than form chemical bonds [89]. Therefore, the process can be described as physicochemical adsorption, dominated by physisorption with minor contributions from specific interactions such as hydrogen bonding [90]. The positive entropy change ($\Delta S^\circ = 0.1141$ kJ/mol/K) indicates increased disorder at the solid–liquid interface, possibly due to desolvation of dye molecules or reorganization of surface functional groups during adsorption [91]. Overall, the thermodynamic analysis indicates that the adsorption of CR onto this DP-AC carbon material is spontaneous and endothermic, mainly governed by physisorption, a behavior commonly seen with bio-based adsorbents.

4 Conclusion

The findings of this work clearly demonstrate the effectiveness of our DP-AC derived from *Phoenix dactylifera* (Ghars variety) pits in removing CR dye from aqueous solutions. The phosphoric acid activation produced a material with high adsorption capacity, reaching 156.04 mg/L. The adsorption process followed the Langmuir isotherm, while kinetic data aligned with the pseudo-second-order model, highlighting the role of surface interactions. Thermodynamic parameters confirmed that the adsorption is spontaneous, endothermic, and governed by physisorption mechanisms. These results highlight the strong potential of DP-AC as efficient, low-cost, and eco-friendly adsorbents for the treatment of dye-contaminated wastewater. Future studies should assess the economic viability and life cycle sustainability of DP-AC, investigate its effectiveness in continuous flow and multi-contaminant systems, and incorporate additional physicochemical evaluations to support its large-scale and sustainable application in wastewater treatment.

Data availability statement The datasets generated during the current study are available from the corresponding author.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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